Convex KKT First-Order Conditions and Support Vector Machines

We return to the first-order conditions of Chapter 12 and ask what more can be deduced when the problem is convex. Our starting point is the variational inequality from Theorem 4.5. We shall connect it to KKT multipliers, keeping track of where convexity and a constraint qualification enter the argument. After separating optimality from the questions of existence and uniqueness, we shall develop hard-margin and soft-margin support vector machines (SVMs) as applications: first from their geometry, then through a convex model, and finally through the interpretation of its KKT conditions.

Learning objectives

After this chapter, you should be able to:

- recognize the canonical smooth convex program;
- relate feasible-set and tangent-cone variational inequalities to KKT;
- distinguish certificates, multiplier existence, primal attainment, and uniqueness;
- determine when a convex minimizer is unique;
- model hard-margin and soft-margin SVM and interpret support vectors.

14.1 Canonical smooth constrained convex programs

Let $\mathcal{D} \subseteq \mathbb{R}^n$ be an open convex function domain. The canonical problem is

$$\min_{\boldsymbol{x}\in\mathcal{C}} f(\boldsymbol{x}), \qquad \mathcal{C} := \{\boldsymbol{x} \in \mathcal{D} : g_i(\boldsymbol{x}) = 0 \ (i = 1, \dots, m),\ h_j(\boldsymbol{x}) \le 0 \ (j = 1, \dots, p)\}, \tag{14.1}$$

where f and every h_j are continuously differentiable and convex on $\mathcal{D}$, while $g_i(\boldsymbol{x}) = \boldsymbol{a}_i^{\mathsf{T}}\boldsymbol{x} - b_i$ is affine. Collect the equalities as $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$.

To check the geometry of this feasible set, consider its constraints one at a time. Each inequality defines a convex sublevel set by Proposition 3.4, and each affine equality defines the intersection of two affine half-spaces. Intersecting these sets with the convex domain preserves convexity by Proposition 3.1. Thus $\mathcal{C}$ is convex.

Warning

Convexity of an equality function does not settle convexity of its zero set. Both x^2 and $x^2 - 1$ are convex, but their zero sets are respectively the convex singleton $\{0\}$ and the nonconvex set $\{-1, 1\}$. Affine equalities are the *canonical sufficient assumption*; they are not a necessary property of every possible representation of a convex feasible set.

Important note: Why the affine representation matters

With affine g_i and $\boldsymbol{\nu} \geq \mathbf{0}$, the function $\boldsymbol{x} \mapsto L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$ is convex, regardless of the signs of $\boldsymbol{\lambda}$. A nonlinear equality need not preserve this property. The convex-set problem $\min x$ subject to $x^2 = 0$, for instance, has the unique global minimizer zero but no KKT multiplier, since $1 + \lambda(2 \cdot 0) \neq 0$.

Use the active set, Lagrangian, KKT point, and LICQ from Chapter 12. The KKT first-order equations at a feasible point are

$$\nabla f(\boldsymbol{x}^*) + \boldsymbol{A}^{\mathsf{T}}\boldsymbol{\lambda}^* + \sum_{j=1}^{p} \nu_j^* \nabla h_j(\boldsymbol{x}^*) = \mathbf{0}, \qquad \boldsymbol{\nu}^* \geq \mathbf{0}, \qquad \nu_j^* h_j(\boldsymbol{x}^*) = 0. \tag{14.2}$$

Primal feasibility remains part of the full system. Calling these conditions FONC emphasizes their *necessary-condition role*; for convex problems the same conditions are also sufficient, as will be shown shortly.

14.2 The variational inequality and convex KKT first-order characterization

For a nonempty convex feasible set $\mathcal{C}$ and $\boldsymbol{x}^* \in \mathcal{C}$, Theorem 4.5 gives the characterization

$$\boldsymbol{x}^* \in \arg\min_{\boldsymbol{x} \in \mathcal{C}} f(\boldsymbol{x}) \quad \Longleftrightarrow \quad \nabla f(\boldsymbol{x}^*)^{\mathsf{T}}(\boldsymbol{x} - \boldsymbol{x}^*) \geq 0 \quad (\boldsymbol{x} \in \mathcal{C}). \tag{14.3}$$

The variational inequality tests every feasible point. For the canonical problem, we seek an algebraic certificate involving only the gradients at $\boldsymbol{x}^*$ and the KKT multipliers. We establish the two directions in turn:

$$\begin{aligned} \text{Variational inequality} + \text{LICQ} &\implies \text{KKT first-order conditions}, \\ \text{KKT first-order conditions} &\implies \text{Variational inequality}. \end{aligned}$$

The second implication needs no constraint qualification.

14.2.1 Variational inequality and LICQ imply the KKT first-order conditions

Suppose that the feasible point $\boldsymbol{x}^*$ satisfies the variational inequality and LICQ. Recall from Chapter 12 that the linearized cone is

$$T_{\text{lin}}(\boldsymbol{x}^*) = \{\boldsymbol{d} : \boldsymbol{A}\boldsymbol{d} = \mathbf{0},\ \nabla h_j(\boldsymbol{x}^*)^{\mathsf{T}}\boldsymbol{d} \leq 0\ (j \in J(\boldsymbol{x}^*))\}.$$

Here $J(\boldsymbol{x}^*)$ is the active inequality set. LICQ gives the exact linearization

$$T_{\mathcal{C}}(\boldsymbol{x}^*) = T_{\text{lin}}(\boldsymbol{x}^*)$$

by Theorem 12.6. Thus every $\boldsymbol{d} \in T_{\text{lin}}(\boldsymbol{x}^*)$ has a tangent sequence

$$\boldsymbol{x}^{(k)} \in \mathcal{C}, \qquad \boldsymbol{x}^{(k)} \to \boldsymbol{x}^*, \qquad t^{(k)} \downarrow 0, \qquad \frac{\boldsymbol{x}^{(k)} - \boldsymbol{x}^*}{t^{(k)}} \to \boldsymbol{d}.$$

Apply the variational inequality to each $\boldsymbol{x}^{(k)}$, divide by $t^{(k)} > 0$, and pass to the limit:

$$\nabla f(\boldsymbol{x}^*)^\mathsf{T}\left(\frac{\boldsymbol{x}^{(k)} - \boldsymbol{x}^*}{t^{(k)}}\right) \geq 0 \quad \Longrightarrow \quad \nabla f(\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{d} \geq 0.$$

Since $\boldsymbol{d}$ was arbitrary, we have

$$\nabla f(\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{d} \geq 0 \qquad (\boldsymbol{d} \in T_{\text{lin}}(\boldsymbol{x}^*)). \tag{14.4}$$

This rules out the descent-direction alternative in Farkas' lemma (Lemma 12.8). Equivalently, it says $-\nabla f(\boldsymbol{x}^*) \in T_{\text{lin}}(\boldsymbol{x}^*)^\circ$, with polar convention $\boldsymbol{v}^\mathsf{T}\boldsymbol{d} \leq 0$. The Farkas-type representation proved in Proposition 12.9 gives

$$-\nabla f(\boldsymbol{x}^*) = \boldsymbol{A}^\mathsf{T}\boldsymbol{\lambda}^* + \sum_{j \in J(\boldsymbol{x}^*)} \nu_j^* \nabla h_j(\boldsymbol{x}^*), \qquad \nu_j^* \geq 0.$$

Set $\nu_j^* = 0$ on inactive indices. We obtain stationarity, dual feasibility, and complementarity; primal feasibility was assumed. Hence $\boldsymbol{x}^*$ is a KKT point. LICQ was used to identify the tangent and linearized cones. The Farkas representation itself does not require LICQ.

14.2.2 The KKT first-order conditions imply variational inequality

Now suppose that $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ is a KKT triple. For any $\boldsymbol{d} \in T_{\text{lin}}(\boldsymbol{x}^*)$, complementarity makes all inactive multipliers zero, and stationarity together with $\boldsymbol{A}\boldsymbol{d} = \boldsymbol{0}$ gives

$$\nabla f(\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{d} = -\sum_{j \in J(\boldsymbol{x}^*)} \nu_j^* \nabla h_j(\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{d} \geq 0.$$

Each term is nonnegative because $\nu_j^* \geq 0$ and $\nabla h_j(\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{d} \leq 0$. This proves Equation (14.4). Together with the Farkas argument above, it shows that KKT and the linearized-cone inequality are equivalent at a feasible point, without LICQ.

To obtain the variational inequality on $\mathcal{C}$, first use the always-valid inclusion $T_{\mathcal{C}}(\boldsymbol{x}^*) \subseteq T_{\text{lin}}(\boldsymbol{x}^*)$ from Proposition 12.5. The gradient inner product is therefore nonnegative on every tangent direction. Next take any $\boldsymbol{x} \in \mathcal{C}$ and set $\boldsymbol{d} = \boldsymbol{x} - \boldsymbol{x}^*$. Convexity keeps the segment from $\boldsymbol{x}^*$ to $\boldsymbol{x}$ feasible, so

$$\boldsymbol{x}^{(k)} = \boldsymbol{x}^* + \frac{\boldsymbol{d}}{k} \in \mathcal{C}, \qquad \boldsymbol{x}^{(k)} \to \boldsymbol{x}^*, \qquad t^{(k)} = \frac{1}{k}.$$

The difference quotient is exactly

$$\frac{\boldsymbol{x}^{(k)} - \boldsymbol{x}^*}{t^{(k)}} = \boldsymbol{d}.$$

Thus $\boldsymbol{d} \in T_{\mathcal{C}}(\boldsymbol{x}^*)$, and

$$0 \leq \nabla f(\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{d} = \nabla f(\boldsymbol{x}^*)^\mathsf{T}(\boldsymbol{x} - \boldsymbol{x}^*).$$

Since $\boldsymbol{x}$ was arbitrary, this is the variational inequality. No LICQ was used: cone inclusion and convexity of the feasible set suffice.

The tangent-sequence and feasible-segment arguments also establish, for a convex $\mathcal{C}$, the equivalence

$$\nabla f(\boldsymbol{x}^*)^\mathsf{T}(\boldsymbol{x} - \boldsymbol{x}^*) \geq 0 \quad (\boldsymbol{x} \in \mathcal{C}) \quad \Longleftrightarrow \quad \nabla f(\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{d} \geq 0 \quad (\boldsymbol{d} \in T_{\mathcal{C}}(\boldsymbol{x}^*)). \tag{14.5}$$

For the forward direction, only the tangent-sequence limit is needed; the reverse direction uses convex feasible segments.

Important note: The roles of LICQ and convexity

LICQ supplies the exact linearization needed for variational inequality to imply KKT. For KKT to imply variational inequality, the inclusion $T_{\mathcal{C}} \subseteq T_{\text{lin}}$ is enough, and convexity makes every feasible displacement a tangent direction. Convexity of the objective then makes variational inequality equivalent to global optimality by Theorem 4.5. The next theorem collects these implications.

14.2.3 Formal statement

Theorem 14.1 (Convex KKT first-order characterization)**.** *Every KKT point of Equation (14.1) is a global minimizer, without any constraint qualification. At a feasible point $\boldsymbol{x}^*$ where LICQ holds,*

$$\boldsymbol{x}^* \text{ is globally optimal} \iff \boldsymbol{x}^* \text{ is a KKT point.}$$

When these equivalent conditions hold, the KKT multiplier pair is unique.

Proof. By Section 14.2.2, every KKT point satisfies the variational inequality, without any constraint qualification. Then Theorem 4.5 proves global optimality.

Conversely, a global minimizer satisfies the variational inequality by Theorem 4.5. Under LICQ, Section 14.2.1 supplies a KKT multiplier pair; Proposition 12.2 gives its uniqueness. □

Important note: No constraint qualification for sufficiency

The theorem begins with a KKT certificate already available. A constraint qualification is needed to guarantee that multipliers exist at a minimizer, not to verify the certificate once it has been found.

Example 14.1 (A convex KKT calculation)**.** Consider $\min(x_1^2+x_2^2)/2$ subject to $1-x_1-x_2 \le 0$. Stationarity gives $x_1 = x_2 = \nu$. If the inequality were inactive, complementarity would give $\nu = 0$ and hence $\boldsymbol{x} = \mathbf{0}$, which is infeasible. Thus it is active and $x_1 + x_2 = 1$, giving $\boldsymbol{x}^* = (1/2, 1/2)^\mathsf{T}$ and $\nu^* = 1/2$. These values satisfy primal feasibility, dual feasibility, and complementarity. Convex KKT sufficiency certifies global optimality; LICQ holds because the active gradient is $(-1,-1)^\mathsf{T} \neq \mathbf{0}$.

14.2.4 Alternative proofs and affine-constraint necessity

A supporting-hyperplane proof

We can also compare the objective at a KKT point directly with its value at an arbitrary feasible point. The supporting-hyperplane inequality from Theorem 4.4 states that a differentiable convex function ϕ on the open convex domain $\mathcal{D}$ satisfies

$$\phi(\boldsymbol{y}) \ge \phi(\boldsymbol{x}) + \nabla\phi(\boldsymbol{x})^\mathsf{T}(\boldsymbol{y} - \boldsymbol{x}) \qquad (\boldsymbol{x}, \boldsymbol{y} \in \mathcal{D}). \tag{14.6}$$

Fix a KKT triple $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ and let $\boldsymbol{x}$ be any feasible point. Applying Equation (14.6) to f gives

$$\begin{aligned} f(\boldsymbol{x}) - f(\boldsymbol{x}^*) &\geq \nabla f(\boldsymbol{x}^*)^{\mathsf{T}}(\boldsymbol{x} - \boldsymbol{x}^*) \\ &= -\sum_{j=1}^{p} \nu_j^* \nabla h_j(\boldsymbol{x}^*)^{\mathsf{T}}(\boldsymbol{x} - \boldsymbol{x}^*) - (\boldsymbol{A}^{\mathsf{T}}\boldsymbol{\lambda}^*)^{\mathsf{T}}(\boldsymbol{x} - \boldsymbol{x}^*), \end{aligned}$$

where stationarity was used. The equality term is zero because both points satisfy $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$. Convexity of h_j gives

$$\nabla h_j(\boldsymbol{x}^*)^{\mathsf{T}}(\boldsymbol{x} - \boldsymbol{x}^*) \leq h_j(\boldsymbol{x}) - h_j(\boldsymbol{x}^*).$$

Since $\nu_j^* \geq 0$,

$$\begin{aligned} f(\boldsymbol{x}) - f(\boldsymbol{x}^*) &\geq -\sum_j \nu_j^* \big(h_j(\boldsymbol{x}) - h_j(\boldsymbol{x}^*)\big) \\ &= -\sum_j \nu_j^* h_j(\boldsymbol{x}) \geq 0. \end{aligned}$$

The equality uses complementary slackness, and the last inequality uses $h_j(\boldsymbol{x}) \leq 0$ and $\nu_j^* \geq 0$. Thus no feasible point has smaller objective value. The comparison works for every feasible $\boldsymbol{x}$ because the affine tangent function in Equation (14.6) underestimates ϕ throughout its convex domain. This is where a stationarity equation at one point becomes a global optimality certificate.

A Lagrangian proof

For another proof, fix the multipliers at a KKT triple and investigate the function $\boldsymbol{x} \mapsto L(\boldsymbol{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ on $\mathcal{D}$. The Lagrangian is the sum of the convex function f, the convex functions $\nu_j^* h_j$ with nonnegative coefficients, and affine functions $\lambda_i^* g_i$; hence it is convex. KKT stationarity says its gradient vanishes at $\boldsymbol{x}^*$. Applying Equation (14.6) to the Lagrangian gives

$$L(\boldsymbol{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \geq L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \qquad (\boldsymbol{x} \in \mathcal{D}).$$

For feasible $\boldsymbol{x}$, all equalities vanish and $\sum_j \nu_j^* h_j(\boldsymbol{x}) \leq 0$. At $\boldsymbol{x}^*$, every summand is zero by complementary slackness. Consequently,

$$L(\boldsymbol{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \leq f(\boldsymbol{x}) \quad \text{for every feasible } \boldsymbol{x}, \qquad L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = f(\boldsymbol{x}^*).$$

Following these comparisons in order gives

$$f(\boldsymbol{x}) \geq L(\boldsymbol{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \geq L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = f(\boldsymbol{x}^*).$$

Thus the Lagrangian provides a lower bound that meets the objective at $\boldsymbol{x}^*$, which proves global optimality.

Affine constraints need no LICQ for necessity

The necessity argument in Section 14.2.1 used LICQ to replace the tangent cone by the linearized cone. For affine constraints we can check this equality directly: a direction satisfying the active linear inequalities generates a short feasible ray. We shall use this observation to obtain multiplier existence even when the active normals are dependent.

Theorem 14.2 (KKT first-order necessity for affine constraints)**.** *Let f be differentiable and let*

$$P := \{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{B}\boldsymbol{x} \le \boldsymbol{c}\}$$

be nonempty. If $\boldsymbol{x}^*$ is a local minimizer of f over P, then KKT multipliers exist at $\boldsymbol{x}^*$, even if the active affine-constraint normals are linearly dependent. If f is convex, these KKT first-order conditions are also sufficient for global optimality.

Proof. Let the jth row of $\boldsymbol{B}$ be $\boldsymbol{a}_j^\mathsf{T}$, and let $I := \{j : \boldsymbol{a}_j^\mathsf{T}\boldsymbol{x}^* = c_j\}$ denote the active row indices. For affine constraints, the tangent and linearized cones agree without any constraint qualification:

$$T_P(\boldsymbol{x}^*) = \{\boldsymbol{d} : \boldsymbol{A}\boldsymbol{d} = \boldsymbol{0},\ \boldsymbol{a}_j^\mathsf{T}\boldsymbol{d} \le 0\ (j \in I)\}.$$

Indeed, for a tangent sequence $\boldsymbol{x}^{(k)} \in P$ and scales $t^{(k)} \downarrow 0$, divide $\boldsymbol{A}(\boldsymbol{x}^{(k)} - \boldsymbol{x}^*) = \boldsymbol{0}$ and $\boldsymbol{a}_j^\mathsf{T}(\boldsymbol{x}^{(k)} - \boldsymbol{x}^*) \le 0$ for $j \in I$ by $t^{(k)} > 0$ and take limits. This proves the inclusion from left to right. For the converse, if $\boldsymbol{d}$ satisfies the displayed relations, then $\boldsymbol{x}^* + t\boldsymbol{d}$ preserves the equalities and all active inequalities for every $t \ge 0$; each inactive residual is negative at $t = 0$ and remains negative for all sufficiently small $t > 0$ by continuity. Because there are finitely many such residuals, one positive interval works for all of them. A sequence on this feasible ray therefore realizes $\boldsymbol{d}$.

The tangent-cone first-order condition now gives $\nabla f(\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{d} \ge 0$ for every vector in the displayed cone. Applying the cone alternative Lemma 12.8 to $-\nabla f(\boldsymbol{x}^*)$ and the active affine normals yields

$$-\nabla f(\boldsymbol{x}^*) = \boldsymbol{A}^\mathsf{T}\boldsymbol{\lambda}^* + \sum_{j \in I} \nu_j^* \boldsymbol{a}_j, \qquad \nu_j^* \ge 0.$$

Set $\nu_j^* = 0$ for inactive indices. This is stationarity together with dual feasibility and complementarity; primal feasibility is already known. Hence KKT multipliers exist. When f is convex, all inequality functions are affine and Theorem 14.1 proves global optimality. □

14.3 Existence, uniqueness, and second-order certificates

14.3.1 A certificate is not an existence theorem

Before using KKT equations to solve a problem, we should distinguish two existence questions: whether the primal minimum is attained, and whether an attained minimizer admits multipliers. The following examples examine them separately.

Warning

Consider the canonical convex problem

$$\min_{x_1, x_2} e^{x_1} \quad \text{subject to} \quad x_2 = 0.$$

Here $\nabla g = (0, 1)^\mathsf{T} \ne \boldsymbol{0}$, so LICQ holds everywhere on its feasible set. However, $e^{x_1} > 0$ for every finite x_1, whereas $e^{x_1} \to 0$ as $x_1 \to -\infty$. Its infimum is therefore zero and is never attained. Correspondingly, the first stationarity equation $e^{x_1} = 0$ has no solution. Thus even LICQ everywhere does not supply a primal minimizer or a KKT point. The correct existence statement is: *an attained minimizer satisfying LICQ admits KKT multipliers.*

Next consider what can happen when the minimum is attained.

Warning

For the canonical convex problem $\min x$ subject to $x^2 \le 0$, feasibility forces $x = 0$, so zero is the unique global minimizer. At this point the constraint gradient vanishes, and stationarity would require $1 + \nu(2 \cdot 0) = 0$, which is impossible. Here $T_{\mathcal{C}}(0) = \{0\}$, $T_{\text{lin}}(0) = \mathbb{R}$, and LICQ fails. Hence convexity and primal attainment alone do not guarantee a multiplier. By contrast, dependent affine constraints still admit multipliers at a minimizer by Theorem 14.2. Failure of LICQ is not, by itself, a proof of failure of KKT necessity or of a duality gap.

14.3.2 Global and local uniqueness

Once a minimizer has been found, we can ask whether another one is possible.

Uniqueness from strict convexity of objective function. The first criterion to check is strict convexity of f on the feasible set: Theorem 4.1 then gives at most one minimizer. The phrase "at most one" is essential; this criterion does not establish attainment. To see what can be lost under ordinary convexity, consider the next example.

Example 14.2 (A continuum of KKT points)**.** For $\min x_1^2$ subject to $x_2^2 - 1 \le 0$, the objective is nonnegative and equals zero precisely when $x_1 = 0$. The constraint then permits every $x_2 \in [-1, 1]$, so the whole segment $\{(0, x_2) : -1 \le x_2 \le 1\}$ is globally optimal. At each point of this segment the objective gradient vanishes; choosing $\nu = 0$ therefore satisfies stationarity and complementarity. LICQ holds at every such point: there are no active constraints when $|x_2| < 1$, and the sole active gradient $(0, 2x_2)^\mathsf{T}$ is nonzero when $|x_2| = 1$. Thus even convexity and LICQ together do not imply uniqueness.

Uniqueness from SOSC. For twice continuously differentiable canonical convex data, we can instead investigate a KKT point using the second-order sufficient condition from Chapter 13. Suppose a KKT triple satisfies SOSC. Convex KKT sufficiency first gives global minimality, while Theorem 13.11 gives strict local minimality. To relate these two conclusions, suppose $\boldsymbol{y} \neq \boldsymbol{x}^*$ were another global minimizer. Convexity makes the whole segment feasible, and for every $t \in [0, 1]$,

$$f((1-t)\boldsymbol{x}^* + t\boldsymbol{y}) \le (1-t)f(\boldsymbol{x}^*) + tf(\boldsymbol{y}) = f(\boldsymbol{x}^*).$$

Global minimality makes this an equality. Arbitrarily small positive t produce distinct feasible points arbitrarily near $\boldsymbol{x}^*$ with the same value, contradicting strict local minimality. Thus the KKT point is the unique global minimizer. Neither step of this argument requires LICQ.

Strict convexity and SOSC are distinct from each other To compare SOSC with strict convexity, first examine the curvature already supplied by the canonical assumptions. For twice continuously differentiable data, affine equalities have zero Hessian, so

$$\boldsymbol{H}_L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = \boldsymbol{H}_f(\boldsymbol{x}^*) + \sum_j \nu_j^* \boldsymbol{H}_{h_j}(\boldsymbol{x}^*).$$

By Theorem 5.5, convexity gives $\boldsymbol{H}_f \succeq 0$ and $\boldsymbol{H}_{h_j} \succeq 0$, and dual feasibility gives $\nu_j^* \geq 0$. Therefore $\boldsymbol{H}_L \succeq 0$ on the entire space: its quadratic form is a sum of nonnegative quadratic forms. In particular, SONC is automatic at a convex KKT triple and supplies no additional filter for this problem class.

Example 14.3 (Strict convexity need not imply SOSC)**.** Consider

$$\min_{x_1,x_2} x_1^4 + x_2^2 \quad \text{subject to} \quad x_2 = 0.$$

The objective is strictly convex: each scalar derivative $4x_1^3$ and $2x_2$ is strictly increasing, and for two distinct vectors at least one of the corresponding scalar strict convex inequalities is strict. On the feasible line $x_2 = 0$, minimizing the objective reduces to minimizing x_1^4. Hence the origin is the unique minimizer. Stationarity there gives $\lambda^* = 0$, and the linearized equality requires $d_2 = 0$. Since the objective gradient also vanishes, the critical cone is $\{(d_1, 0) : d_1 \in \mathbb{R}\}$. Computing $\boldsymbol{H}_L(\mathbf{0}, 0) = \operatorname{diag}(0, 2)$ now gives zero quadratic form on every horizontal critical direction, so SOSC fails. Strict convexity ensures uniqueness, not positive quadratic curvature; here the fourth-order term determines the minimum.

A sufficient curvature check. To obtain a convenient sufficient test for a twice continuously differentiable canonical convex problem, compare objective curvature with Lagrangian curvature at a KKT triple. For every critical direction,

$$\boldsymbol{d}^\mathsf{T} \boldsymbol{H}_L \boldsymbol{d} = \boldsymbol{d}^\mathsf{T} \boldsymbol{H}_f(\boldsymbol{x}^*) \boldsymbol{d} + \sum_j \nu_j^* \boldsymbol{d}^\mathsf{T} \boldsymbol{H}_{h_j}(\boldsymbol{x}^*) \boldsymbol{d} \geq \boldsymbol{d}^\mathsf{T} \boldsymbol{H}_f(\boldsymbol{x}^*) \boldsymbol{d}.$$

Consequently, if $\boldsymbol{d}^\mathsf{T} \boldsymbol{H}_f(\boldsymbol{x}^*) \boldsymbol{d} > 0$ for every nonzero $\boldsymbol{d} \in C_{\text{crit}}(\boldsymbol{x}^*)$, the same strict inequality holds for the Lagrangian quadratic form. This proves SOSC. Positive definiteness $\boldsymbol{H}_f(\boldsymbol{x}^*) \succ 0$ is one sufficient way to meet this test.

We can also check what strong convexity supplies. If f is strongly convex with parameter $\mu > 0$, apply Equation (10.15) to $\boldsymbol{x}^*$ and $\boldsymbol{x}^* + t\boldsymbol{d}$, then use the second-order expansion at $\boldsymbol{x}^*$. After subtracting the constant and linear terms, dividing by $t^2/2$, and letting $t \to 0$, it gives $\boldsymbol{d}^\mathsf{T} \boldsymbol{H}_f(\boldsymbol{x}^*) \boldsymbol{d} \geq \mu \|\boldsymbol{d}\|^2$. Thus a strongly convex objective satisfies the same curvature test. The next example examines a different source of positive Lagrangian curvature: the inequality constraints themselves.

Example 14.4 (SOSC need not require a strictly convex objective)**.** For

$$\min_{x_1,x_2} x_2 \quad \text{subject to} \quad x_1^2 - x_2 \leq 0,$$

feasibility gives $x_2 \geq x_1^2 \geq 0$, with equality only at the origin. The objective is linear, not strictly convex. At the origin stationarity gives $\nu^* = 1$. The linearized cone is $\{d_2 \geq 0\}$ and the critical cone is $\{d_2 = 0\}$. The Lagrangian at $\nu^* = 1$ is x_1^2, so

$$\boldsymbol{H}_L = \operatorname{diag}(2, 0), \qquad \boldsymbol{d}^\mathsf{T} \boldsymbol{H}_L \boldsymbol{d} = 2d_1^2 > 0 \quad (\boldsymbol{d} \in C_{\text{crit}}(\mathbf{0}) \setminus \{\mathbf{0}\}).$$

Thus the curved loaded inequality supplies the SOSC curvature even though the objective is not strictly convex.

14.4 Workflows for optimality analysis

The preceding arguments suggest the following order for analyzing a new convex problem. It keeps the search for a candidate separate from the justifications for optimality, existence, and uniqueness.

1. Confirm convexity of f, h_j, affinity of g_i, and primal feasibility.
2. Write the full KKT first-order system, including multiplier signs and complementarity. Use active-set cases when convenient.
3. Solve for feasible KKT triples. Every triple found is a global certificate, whether or not LICQ holds.
4. If no triple is found, justify multiplier necessity before rejecting a minimizer. Distinguish infeasibility, nonattainment, failed qualifications, and an unsuccessful calculation.
5. For uniqueness, use strict convexity on the feasible set or a KKT SOSC certificate. Uniqueness is a separate issue from global optimality.

Important note: Writing the equations is always allowed

LICQ is a theorem hypothesis, not a prerequisite for writing or attempting to solve KKT equations. An unqualified convex problem can still have KKT points, and every one is globally optimal. LICQ at a specified feasible point guarantees the reverse implication there; it does not guarantee an optimum exists somewhere.

The convex KKT certificate completes the convex branch of the workflow in Figure 13.4. In a canonical convex problem, SONC is automatic; strict convexity or SOSC can additionally establish uniqueness, as explained above.

14.5 Hard-margin support vector machines

Let $(\boldsymbol{x}_i, y_i)$, $i = 1, \dots, N$, be a finite labeled data set with $\boldsymbol{x}_i \in \mathbb{R}^n$ and $y_i \in \{-1, +1\}$. For $\boldsymbol{w} \neq \mathbf{0}$, the hyperplane $\boldsymbol{w}^\mathsf{T}\boldsymbol{x} + b = 0$ predicts the sign of $\boldsymbol{w}^\mathsf{T}\boldsymbol{x} + b$. When several hyperplanes separate the labels correctly, we need a criterion for choosing between them. We shall measure the empty band around a separator and maximize its width. The geometric model will then lead, by rescaling the separating variables, to a canonical convex program.

Throughout the geometric discussion, assume both labels occur. We call the data *strictly linearly separable* if there are $(\boldsymbol{w}, b)$ such that

$$y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) > 0 \qquad (i = 1{:}N).$$

We assume strict separability while developing the hard-margin geometry. If $\boldsymbol{w} = \mathbf{0}$, the score is the same constant b on every observation and cannot be strictly positive on one class and strictly negative on the other. Thus a strict separator has $\boldsymbol{w} \neq \mathbf{0}$. Both labels are essential here for a two-sided margin. Figure 14.1 illustrates how the distance between the two supporting hyperplanes varies with the separator. The algebraic one-class case is handled separately in the existence proof below, but it does not describe a two-sided separating hyperplane with a finite margin width.

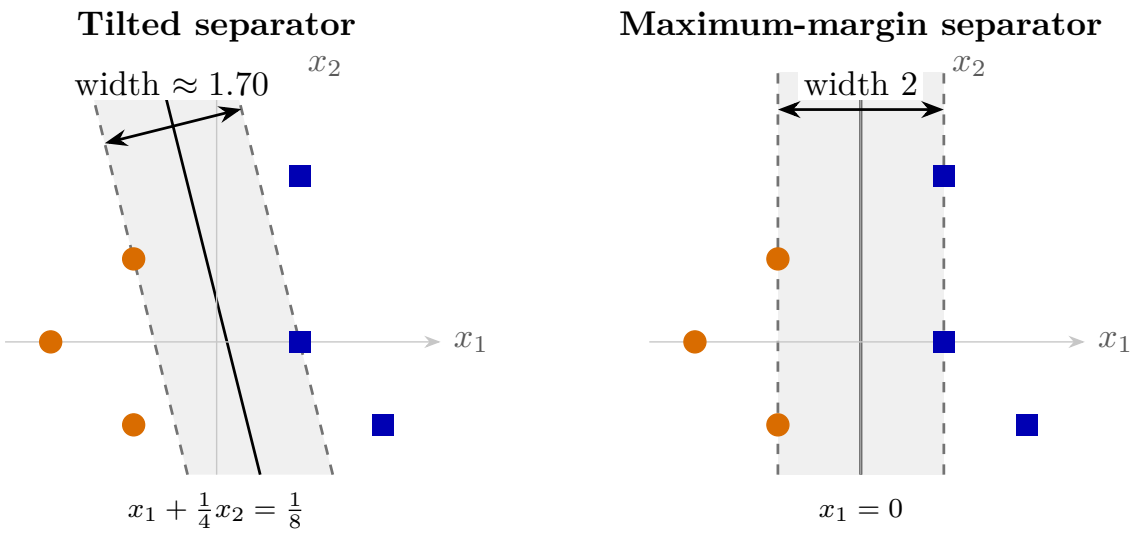


Figure 14.1: A tilted separator and the maximum-margin separator for the six-point data below. The empty-band widths are approximately 1.70 and 2. Each displayed band is balanced between its two supporting planes; optimizing the normal increases the available width.

Distance and margin geometry. To measure a margin, first find the distance from $\boldsymbol{x}_0$ to a level plane $\{\boldsymbol{x} : \boldsymbol{w}^\mathsf{T}\boldsymbol{x} + b = c\}$, where $\boldsymbol{w} \neq \boldsymbol{0}$. Move from $\boldsymbol{x}_0$ in the normal direction by considering $\boldsymbol{x}_0 + t\boldsymbol{w}$. Its residual from the plane is

$$\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_0 + b - c + t\,\|\boldsymbol{w}\|^2 .$$

Setting this residual to zero gives $t = -(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_0 + b - c)/\,\|\boldsymbol{w}\|^2$, so this displacement reaches the plane after travelling distance $|\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_0 + b - c|/\,\|\boldsymbol{w}\|$. To verify that no shorter displacement is possible, let $\boldsymbol{s}$ be any vector for which $\boldsymbol{x}_0 + \boldsymbol{s}$ lies on the plane. Then

$$|\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_0 + b - c| = |\boldsymbol{w}^\mathsf{T}\boldsymbol{s}| \leq \|\boldsymbol{w}\|\,\|\boldsymbol{s}\|$$

by Cauchy–Schwarz. The normal displacement attains this lower bound, and hence the Euclidean distance is

$$\frac{|\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_0 + b - c|}{\|\boldsymbol{w}\|}. \tag{14.7}$$

If $\boldsymbol{x}_0$ lies on the level plane with value c_1, its residual relative to the plane with value c_2 is $c_1 - c_2$. The distance between these parallel planes is therefore $|c_1 - c_2|/\,\|\boldsymbol{w}\|$. In particular, the planes with values $+1$ and -1 are $2/\,\|\boldsymbol{w}\|$ apart.

14.5.1 The geometric maximum-margin model

We can now use the distance formula to choose a margin criterion. Set $c = 0$ and attach the label y_i to the score of observation i. Its signed distance, positive when it is on its correctly labeled side, is

$$\frac{y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b)}{\|\boldsymbol{w}\|}.$$

For a strict separator, the largest centered empty-band half-width is therefore

$$\alpha(\boldsymbol{w}, b) := \min_{1 \leq i \leq N} \frac{y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b)}{\|\boldsymbol{w}\|} > 0.$$

To require a band of half-width at least $\alpha > 0$, impose

$$y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) \geq \alpha \|\boldsymbol{w}\| \qquad (i = 1{:}N).$$

The slab $\{\boldsymbol{x} : |\boldsymbol{w}^\mathsf{T}\boldsymbol{x} + b| < \alpha \|\boldsymbol{w}\|\}$ then contains no observation. Its two boundary planes are each at distance α from the decision plane, so its full width is 2α. Thus the geometric formulation is

$$\begin{aligned} \max_{\alpha,\boldsymbol{w},b} \quad & \alpha \\ \text{subject to} \quad & y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) \geq \alpha \|\boldsymbol{w}\|, \qquad i = 1{:}N, \\ & \alpha > 0, \qquad \boldsymbol{w} \neq \boldsymbol{0}. \end{aligned} \tag{14.8}$$

Strict separability and finiteness give a positive feasible half-width, namely $\alpha(\boldsymbol{w}, b)$ for any strict separator. Maximizing α or 2α gives the same separating hyperplanes.

An observation with equality in a band constraint is a *boundary contact*. A feasible choice of α can be smaller than the largest available half-width, so boundary contacts need not yet exist. At a geometric optimum they are the geometric support vectors discussed below.

Warning

The condition $\boldsymbol{w} \neq \boldsymbol{0}$ must be included in the geometric model. Without it, $\boldsymbol{w} = \boldsymbol{0}$, $b = 0$ satisfies every displayed band inequality for arbitrarily large α, although it defines no separating hyperplane. Also, normalization by $\beta = \alpha \|\boldsymbol{w}\|$ requires $\alpha > 0$: a zero-width band cannot be divided by its zero functional margin.

14.5.2 Normalization to the canonical convex program

The product $\alpha \|\boldsymbol{w}\|$ in the constraints makes the geometric formulation inconvenient to solve directly. To separate its scale from its geometry, set $\beta := \alpha \|\boldsymbol{w}\| > 0$. Since $\boldsymbol{w} \neq \boldsymbol{0}$, this is equivalent to $\alpha = \beta / \|\boldsymbol{w}\|$. The geometric problem becomes

$$\begin{aligned} \max_{\beta,\boldsymbol{w},b} \quad & \frac{\beta}{\|\boldsymbol{w}\|} \\ \text{subject to} \quad & y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) \geq \beta, \qquad i = 1{:}N, \\ & \beta > 0, \qquad \boldsymbol{w} \neq \boldsymbol{0}. \end{aligned}$$

Introduce distinct normalized variables

$$\widetilde{\boldsymbol{w}} := \frac{\boldsymbol{w}}{\beta}, \qquad \widetilde{b} := \frac{b}{\beta}.$$

The constraints become $y_i(\widetilde{\boldsymbol{w}}^\mathsf{T}\boldsymbol{x}_i + \widetilde{b}) \geq 1$, and the geometric half-width is

$$\alpha = \frac{\beta}{\|\boldsymbol{w}\|} = \frac{1}{\|\widetilde{\boldsymbol{w}}\|}.$$

Maximizing this positive reciprocal is equivalent to minimizing $\|\widetilde{\boldsymbol{w}}\|^2/2$. With both labels present, the normalized constraints automatically exclude $\widetilde{\boldsymbol{w}} = \boldsymbol{0}$: positive labels would require $\widetilde{b} \geq 1$, whereas negative labels would require $\widetilde{b} \leq -1$. Thus the normalized problem needs no separate nonzero-normal constraint. After renaming the normalized variables as $(\boldsymbol{w}, b)$, hard-margin SVM is

$$\min_{\boldsymbol{w},b} \frac{1}{2}\|\boldsymbol{w}\|^2 \quad \text{subject to} \quad h_i(\boldsymbol{w}, b) := 1 - y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) \leq 0, \quad i = 1, \ldots, N. \tag{14.9}$$

To check that the new formulation fits our earlier theory, compute the Hessian of its objective with respect to $(\boldsymbol{w}, b)$: it is $\operatorname{diag}(\boldsymbol{I}, 0) \succeq 0$. Every h_i is affine, and there are no equality constraints. Thus Equation (14.9) is a canonical convex program.

Canonical scaling and margin. We should verify that this change of variables preserves the optimization problem in both directions. Keep the assumptions of finite, strictly linearly separable two-class data. A feasible geometric triple has $\beta = \alpha \|\boldsymbol{w}\| > 0$. Division of each constraint by β makes $(\widetilde{\boldsymbol{w}}, \widetilde{b})$ canonically feasible, with $\|\widetilde{\boldsymbol{w}}\| = 1/\alpha$. Conversely, a canonical feasible pair $(\widetilde{\boldsymbol{w}}, \widetilde{b})$ cannot have $\widetilde{\boldsymbol{w}} = \boldsymbol{0}$: its positive-class constraints would require $\widetilde{b} \geq 1$, while its negative-class constraints would require $\widetilde{b} \leq -1$. For any $\beta > 0$, set

$$\boldsymbol{w} = \beta \widetilde{\boldsymbol{w}}, \qquad b = \beta \widetilde{b}, \qquad \alpha = \frac{1}{\|\widetilde{\boldsymbol{w}}\|}.$$

Then $\alpha \|\boldsymbol{w}\| = \beta$, and multiplication of the canonical inequalities by β gives all geometric constraints. Thus both constructions preserve feasibility and relate the objective values by $\|\widetilde{\boldsymbol{w}}\|^2 / 2 = 1/(2\alpha^2)$. The latter is strictly decreasing for $\alpha > 0$, so a better feasible objective in either problem would give a better one in the other. Hence geometric maximizers correspond exactly to canonical minimizers under normalization.

There is still a distinction between satisfying the normalized inequalities and having an observation exactly on a margin plane. To investigate it, take any strict separator and form $\gamma := \min_i y_i(\boldsymbol{w}^\mathsf{T} \boldsymbol{x}_i + b)$. There are finitely many strictly positive signed responses, so $\gamma > 0$. Rescaling by $(\boldsymbol{w}, b) \mapsto (\boldsymbol{w}/\gamma, b/\gamma)$ makes this minimum one. For a canonical feasible pair $\gamma \geq 1$. If $\gamma > 1$, the same rescaling keeps all constraints feasible but decreases the positive objective by the factor $1/\gamma^2$. Consequently, a canonical optimum must have $\gamma = 1$. Finally, the normalized margin planes have values ± 1, so Equation (14.7) gives a full band width $2/\|\boldsymbol{w}\|$. Every canonical feasible pair guarantees such an empty centered band; at a canonical optimum the minimum signed response is exactly one, and the band achieves the largest geometric width.

Important note: Geometric margin and functional scale

Multiplying $(\boldsymbol{w}, b)$ by a positive constant leaves both the decision hyperplane and its geometric distances unchanged. Normalization is not needed to compare these distances. It fixes the arbitrary *functional* scale $\beta = \alpha \|\boldsymbol{w}\|$ so that the same geometric optimization has a convenient convex quadratic formulation.

14.5.3 KKT system and support vectors

To determine an optimal separator and interpret the observations that influence it, derive the KKT system of Equation (14.9). Let $\nu_i \geq 0$ be the multiplier of the ith margin constraint. The Lagrangian is

$$L(\boldsymbol{w}, b, \boldsymbol{\nu}) = \frac{1}{2} \|\boldsymbol{w}\|^2 + \sum_{i=1}^{N} \nu_i \left[1 - y_i(\boldsymbol{w}^\mathsf{T} \boldsymbol{x}_i + b)\right]. \tag{14.10}$$

First differentiate with respect to the separating variables:

$$\nabla_{\boldsymbol{w}} L = \boldsymbol{w} - \sum_i \nu_i y_i \boldsymbol{x}_i, \qquad \frac{\partial L}{\partial b} = -\sum_i \nu_i y_i.$$

Setting these derivatives to zero produces the first two equations below. Adjoining the margin inequalities, multiplier signs, and complementary slackness gives the full KKT system:

$$\boldsymbol{w}^* = \sum_{i=1}^{N} \nu_i^* y_i \boldsymbol{x}_i, \tag{14.11}$$

$$\sum_{i=1}^{N} \nu_i^* y_i = 0, \tag{14.12}$$

$$y_i(\boldsymbol{w}^{*\mathsf{T}} \boldsymbol{x}_i + b^*) \geq 1, \qquad i = 1{:}N, \tag{14.13}$$

$$\nu_i^* \left[1 - y_i(\boldsymbol{w}^{*\mathsf{T}} \boldsymbol{x}_i + b^*)\right] = 0, \qquad \nu_i^* \geq 0. \tag{14.14}$$

To understand which observations appear in the normal-vector sum, inspect Equation (14.14). If $\nu_i^* > 0$, complementary slackness forces $1 - y_i(\boldsymbol{w}^{*\mathsf{T}} \boldsymbol{x}_i + b^*) = 0$, so the point lies on a canonical margin plane.

A *geometric support vector* has an active hard-margin constraint: $y_i(\boldsymbol{w}^{*\mathsf{T}} \boldsymbol{x}_i + b^*) = 1$. A *loaded support vector* has $\nu_i^* > 0$. Every loaded support vector is geometric, but a geometric support vector may be unloaded at a chosen KKT multiplier vector.

Terms with $\nu_i^* = 0$ vanish from Equation (14.11), so only the loaded observations contribute directly to the separating normal. Since both classes occur, feasibility gives $\boldsymbol{w}^* \neq \mathbf{0}$; the sum in that equation must therefore contain a positive multiplier. Now separate Equation (14.12) by label:

$$\sum_{\{i:y_i=+1\}} \nu_i^* = \sum_{\{i:y_i=-1\}} \nu_i^*.$$

If either sum were zero, nonnegativity would make every multiplier zero. Thus each class has at least one loaded observation. The equations place no restriction of exactly one per class; multiple geometric or loaded support vectors are possible.

Important note: Two conventions for the term support vector

In this chapter, support vector without a qualifier means a *geometric* support vector, matching an active margin constraint. When interpreting the normal-vector sum, the qualifier *loaded* identifies the positive-multiplier points. The distinction is important for degenerate active constraints.

Warning

The implication from active to loaded can fail in a degenerate data set. A point may lie on a margin plane while its multiplier is zero. This is the same distinction as an active but unloaded wall in Chapter 12.

Recovering the intercept. After finding $\boldsymbol{w}^*$, we can recover b^* from any active margin constraint. Such a constraint is $y_i(\boldsymbol{w}^{*\mathsf{T}} \boldsymbol{x}_i + b^*) = 1$. Multiply by y_i and use $y_i^2 = 1$ to obtain

$$b^* = y_i - \boldsymbol{w}^{*\mathsf{T}} \boldsymbol{x}_i.$$

At an exact KKT solution, every such index i gives the same value. A positive multiplier is not required.

In a numerical solution, small feasibility errors can give slightly different intercept estimates. Averaging $y_i - \boldsymbol{w}^{\mathsf{T}} \boldsymbol{x}_i$ over suitable identified margin points is a common way to reduce this

variation; it does not change the exact formula. Writing $A \neq \varnothing$ for the indices of such reliable near-active observations gives the slides' approximation

$$b \approx \frac{1}{|A|} \sum_{i \in A} (y_i - \boldsymbol{w}^\mathsf{T} \boldsymbol{x}_i).$$

Here A is a sample-index set, distinct from the equality-constraint matrix $\boldsymbol{A}$.

14.5.4 Attainment, uniqueness, and second-order curvature

The KKT equations describe a candidate solution, but we still need to establish that the hard-margin problem has an optimum and determine which parts of the solution are unique. Assume throughout this subsection that Equation (14.9) is feasible.

Attainment. Begin with attainment. If only one label occurs, choose $(\boldsymbol{w}, b) = (\boldsymbol{0}, 1)$ for the positive class or $(\boldsymbol{0}, -1)$ for the negative class. This pair is feasible and its objective is zero. Since the objective is nonnegative, it attains the minimum. Suppose instead that both labels occur. A feasible minimizing sequence $(\boldsymbol{w}^{(k)}, b^{(k)})$ has bounded objective values, hence bounded $\boldsymbol{w}^{(k)}$. Fix one positive index i_+ and one negative index i_-. Their constraints give

$$1 - \boldsymbol{w}^{(k)\mathsf{T}} \boldsymbol{x}_{i_+} \leq b^{(k)} \leq -1 - \boldsymbol{w}^{(k)\mathsf{T}} \boldsymbol{x}_{i_-},$$

so $b^{(k)}$ is bounded. A convergent subsequence has a feasible limit because all constraints are closed, and continuity makes this limit a global minimizer. These two cases prove attainment whenever the hard-margin problem is feasible.

To justify using KKT at this minimizer, note that all constraints are affine. Thus Theorem 14.2 gives multiplier necessity even with dependent active normals, and Theorem 14.1 gives sufficiency. We have therefore obtained an exact KKT characterization of the hard-margin optima without assuming LICQ.

Uniqueness. For uniqueness, consider the normal vector first. If two optima $(\boldsymbol{w}_1, b_1)$ and $(\boldsymbol{w}_2, b_2)$ had $\boldsymbol{w}_1 \neq \boldsymbol{w}_2$, convexity of the constraints would make their midpoint feasible. Strict convexity in $\boldsymbol{w}$ would then give

$$\frac{1}{2} \left\| \frac{\boldsymbol{w}_1 + \boldsymbol{w}_2}{2} \right\|^2 < \frac{1}{2} \left(\frac{1}{2} \|\boldsymbol{w}_1\|^2 + \frac{1}{2} \|\boldsymbol{w}_2\|^2 \right),$$

a value below the minimum. Thus all optima have a common $\boldsymbol{w}^*$.

Next suppose both classes occur, so feasibility forces $\boldsymbol{w}^* \neq \boldsymbol{0}$. For this fixed normal, the constraints on the intercept are exactly

$$\ell \leq b \leq u, \qquad \ell := \max_{\{i : y_i = +1\}} (1 - \boldsymbol{w}^{*\mathsf{T}} \boldsymbol{x}_i), \qquad u := \min_{\{i : y_i = -1\}} (-1 - \boldsymbol{w}^{*\mathsf{T}} \boldsymbol{x}_i).$$

The finite, nonempty classes make both bounds well-defined, and feasibility gives $\ell \leq u$. Suppose $\ell < u$ and choose $b_0 = (\ell + u)/2$. For a positive observation, $\boldsymbol{w}^{*\mathsf{T}} \boldsymbol{x}_i + b_0 > 1$; for a negative observation, $\boldsymbol{w}^{*\mathsf{T}} \boldsymbol{x}_i + b_0 < -1$. Thus every margin constraint is strict, and finiteness gives

$$\gamma := \min_{1 \leq i \leq N} y_i (\boldsymbol{w}^{*\mathsf{T}} \boldsymbol{x}_i + b_0) > 1.$$

The rescaled pair

$$\widehat{\boldsymbol{w}} := \frac{\boldsymbol{w}^*}{\gamma}, \qquad \widehat{b} := \frac{b_0}{\gamma}$$

remains feasible, since $y_i(\widehat{\boldsymbol{w}}^\mathsf{T}\boldsymbol{x}_i + \widehat{b}) \geq 1$ for every i. But

$$\frac{1}{2}\|\widehat{\boldsymbol{w}}\|^2 = \frac{1}{\gamma^2}\frac{1}{2}\|\boldsymbol{w}^*\|^2 < \frac{1}{2}\|\boldsymbol{w}^*\|^2,$$

contradicting optimality. Hence $\ell = u$, so $b^* = \ell = u$ is unique.

We can also examine the intercept through KKT, which connects its uniqueness to support vectors. At one optimum, stationarity gives

$$\boldsymbol{w}^* = \sum_i \nu_i^* y_i \boldsymbol{x}_i, \qquad \sum_i \nu_i^* y_i = 0, \qquad \nu_i^* \geq 0.$$

The first equality and $\boldsymbol{w}^* \neq \boldsymbol{0}$ imply some multiplier is positive. The second then implies positive multipliers in *both* classes. Choose loaded indices i_+, i_-; complementarity gives ${\boldsymbol{w}^*}^\mathsf{T}\boldsymbol{x}_{i_+} + b^* = 1$ and ${\boldsymbol{w}^*}^\mathsf{T}\boldsymbol{x}_{i_-} + b^* = -1$. Any other optimum has the same $\boldsymbol{w}^*$, and feasibility at these two observations forces its intercept to be at least b^* and at most b^*. Hence its intercept equals b^*. This does not make the multiplier representation unique.

Important note: What is unique in hard-margin SVM?

For finite, strictly linearly separable data with both labels, uniqueness refers to the normalized primal parameters $(\boldsymbol{w}^*, b^*)$ in Equation (14.9), not to the KKT triple: different multiplier vectors can certify the same pair, as the worked example below shows. The objective is strictly convex in $\boldsymbol{w}$ but not jointly in $(\boldsymbol{w}, b)$; the two-sided constraints force the feasible intercept interval to collapse at the optimal normal. In the unnormalized geometric model Equation (14.8), set $\alpha^* = 1/\|\boldsymbol{w}^*\|$. Then $(\alpha^*, c\boldsymbol{w}^*, cb^*)$ represents the same optimal band for any $c > 0$, so its coefficient representation is not unique. This hard-margin conclusion does not assert intercept uniqueness for soft-margin SVM introduced later.

Many observations can lie on one margin plane, making their active gradients dependent. Thus LICQ is often stronger than necessary for SVM. The affine-constraint normal-cone argument above gives multiplier existence directly.

Second-order curvature with two classes. To check how this uniqueness appears in the second-order conditions, take an optimal KKT triple for feasible data containing both labels. All constraints are affine, so $\boldsymbol{H}_L = \operatorname{diag}(\boldsymbol{I}, 0)$. Write a critical direction as $(\boldsymbol{d}_w, d_b)$. If $\boldsymbol{d}_w \neq \boldsymbol{0}$, its Lagrangian quadratic form is $\|\boldsymbol{d}_w\|^2 > 0$. If $\boldsymbol{d}_w = \boldsymbol{0}$, the critical-cone representation Equation (13.4) requires zero directional derivative of every loaded constraint. At a loaded index this derivative is $-y_i(\boldsymbol{x}_i^\mathsf{T}\boldsymbol{d}_w + d_b) = -y_i d_b$, so $d_b = 0$. A loaded index exists because $\boldsymbol{w}^* \neq \boldsymbol{0}$ and Equation (14.11) holds. Thus every nonzero critical direction has $\boldsymbol{d}_w \neq \boldsymbol{0}$, which proves SOSC at every optimal KKT triple. Although the objective has zero curvature in the b direction, the loaded constraints exclude a nonzero pure-intercept critical direction. Joint strict convexity of the objective is therefore unnecessary for this conclusion.

Example 14.5 (A complete two-dimensional hard-margin calculation)**.** Take the six

observations

i	1	2	3	4	5	6
$\boldsymbol{x}_i$	$(-1,-1)$	$(-1,1)$	$(1,0)$	$(1,2)$	$(-2,0)$	$(2,-1)$
y_i	-1	-1	$+1$	$+1$	-1	$+1$

and try $\boldsymbol{w}^* = (1,0)^\mathsf{T}$, $b^* = 0$. The signed responses are $(1,1,1,1,2,2)$, so the geometric support vectors are $1,2,3,4$. Complementarity makes $\nu_5 = \nu_6 = 0$. Stationarity in $\boldsymbol{w}$ and b reduces to

$$\nu_1 + \nu_2 + \nu_3 + \nu_4 = 1, \qquad \nu_1 - \nu_2 + 2\nu_4 = 0, \qquad -\nu_1 - \nu_2 + \nu_3 + \nu_4 = 0.$$

To solve these equations, add and subtract the first and third to obtain $\nu_1+\nu_2 = \nu_3+\nu_4 = 1/2$. Put $t = \nu_4$. Then $\nu_3 = 1/2 - t$, and the second equation together with $\nu_1 + \nu_2 = 1/2$ gives $\nu_1 = 1/4 - t$ and $\nu_2 = 1/4 + t$. Nonnegativity imposes $t \geq 0$ and $t \leq 1/4$; these bounds also ensure $\nu_2, \nu_3 \geq 0$. Hence the entire nonnegative multiplier family is

$$\boldsymbol{\nu}^* = \left(\tfrac{1}{4} - t, \tfrac{1}{4} + t, \tfrac{1}{2} - t, t, 0, 0\right)^\mathsf{T}, \qquad 0 \leq t \leq \tfrac{1}{4}.$$

In particular, $t = 0$ gives

$$\boldsymbol{\nu}^* = (1/4, 1/4, 1/2, 0, 0, 0)^\mathsf{T}.$$

All KKT first-order conditions hold, so $\boldsymbol{w}^*, b^*$ are globally optimal. The decision plane is $x_1 = 0$, the margin planes are $x_1 = -1$ and $x_1 = 1$, and the full margin width is 2. The chosen multipliers load points $1,2,3$, while point 4 touches the positive margin but is unloaded. See Figure 14.2 for an illustration.

There are four active constraints in three optimization variables (w_1, w_2, b), so LICQ fails. The affine-constraint theorem still guarantees the exact KKT characterization, and the separating pair is unique although multipliers are not. Finally, compute the critical cone for this example. Zero first-order objective change gives $d_{w_1} = 0$; the first two active constraints give $d_b \leq -|d_{w_2}|$, while the third gives $d_b \geq 0$. Thus $d_b = d_{w_2} = 0$, and the critical cone is $\{\mathbf{0}\}$.

14.6 Soft-margin support vector machines

The hard-margin construction began with strict separability. To extend the model to data for which this assumption fails, we shall first examine a nonseparable configuration and then allow controlled violations of the margin constraints.

14.6.1 When no strict linear separator exists

Example 14.6 (Why a strict separator may not exist)**.** For the four points $(-1,-1), (1,1)$ with label $+1$ and $(-1,1), (1,-1)$ with label -1, the midpoint of each class is the origin; see Figure 14.3. If a strict linear separator existed, averaging the two positive scores would give $b > 0$, whereas averaging the two negative scores would give $b < 0$. This contradiction proves nonseparability. Equivalently, the normalized hard-margin constraints would imply $b \geq 1$ from the positive pair and $b \leq -1$ from the negative pair. The obstruction is the common point of the two class convex hulls.

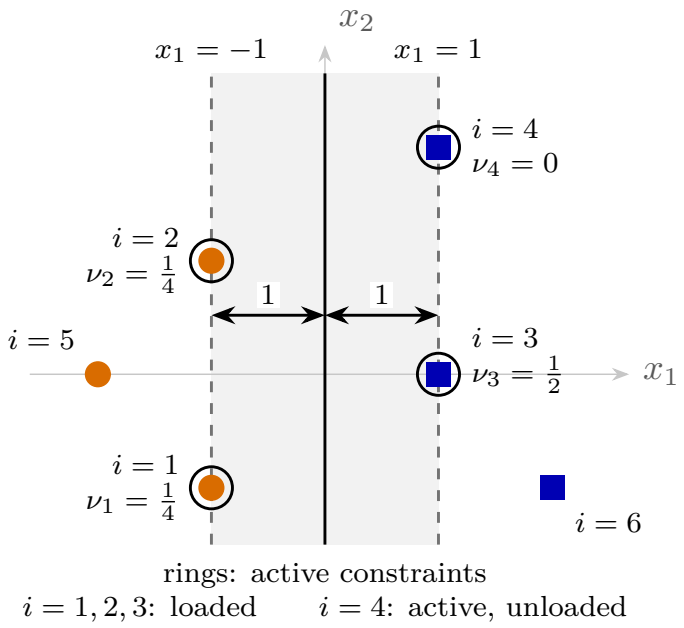


Figure 14.2: The six-point optimal separator, its margin planes, and its four geometric support vectors. Point 4 is active but unloaded for the displayed multiplier choice.

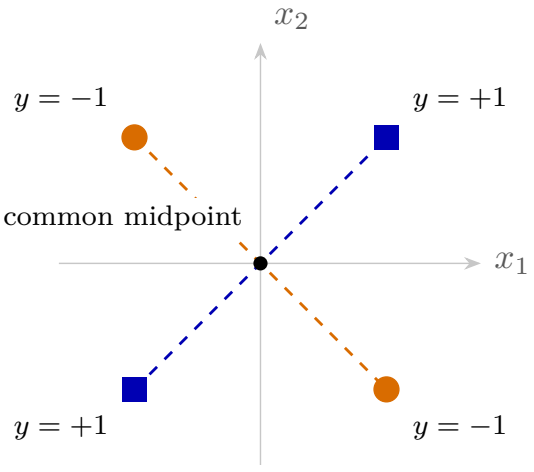


Figure 14.3: The two class segments share the origin, so no strict separating hyperplane exists. This is a feasibility obstruction for hard-margin SVM, not a failure of convexity.

14.6.2 Slack variables and their geometric meaning

For a fixed candidate $(\boldsymbol{w}, b)$, write $s_i = \boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b$. To make a violated normalized margin constraint feasible, introduce a slack $\xi_i \geq 0$ and replace it by $y_i s_i \geq 1 - \xi_i$. Rearranging gives the two lower bounds $\xi_i \geq 1 - y_i s_i$ and $\xi_i \geq 0$. The smallest permitted slack is therefore

$$\xi_i^{\min} = \max\{0, 1 - y_i s_i\}.$$

For the geometric interpretation, take $\xi_i = \xi_i^{\min}$ and assume $\boldsymbol{w} \neq \boldsymbol{0}$. If $\xi_i = 0$, then $y_i s_i \geq 1$, placing the observation on or outside its correct margin plane. If $\xi_i > 0$, minimality of the slack gives $y_i s_i = 1 - \xi_i$. As ξ_i increases through 1, this signed score passes from positive to zero to negative. We therefore obtain the cases

$\xi_i = 0:$	on or outside the correct margin,
$0 < \xi_i < 1:$	inside the margin but correctly classified,
$\xi_i = 1:$	on the decision plane,
$\xi_i > 1:$	misclassified.

See also Figure 14.4. For positive minimal slack, the score differs from its class margin value by ξ_i. Applying Equation (14.7) gives normal-direction distance $\xi_i / \|\boldsymbol{w}\|$ to that margin plane. Thus a slack measures a *functional margin deficit*; the factor $1/\|\boldsymbol{w}\|$ converts it into a Euclidean distance. If $\boldsymbol{w} = \boldsymbol{0}$, this geometric distance interpretation is unavailable; the algebraic soft-margin model below will still allow this case.

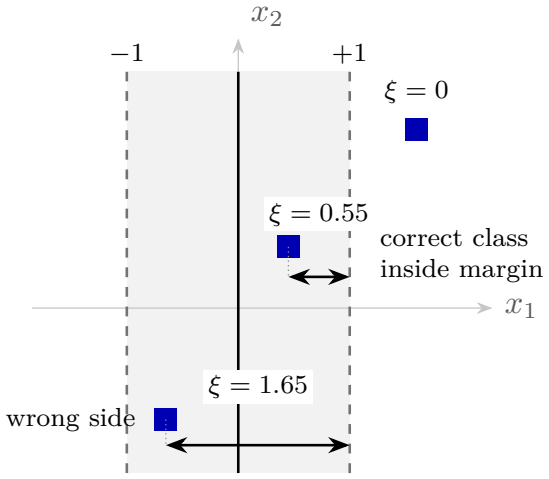


Figure 14.4: Slack geometry for the illustrative pair $\boldsymbol{w} = (1, 0)^\mathsf{T}$, $b = 0$, not a claimed soft-margin optimizer. All three displayed observations have label +1; their minimal slacks are 0, 0.55, and 1.65.

14.6.3 The convex soft-margin model

To allow margin violations while controlling their size, choose a penalty parameter $C > 0$ and solve

$$\min_{\boldsymbol{w}, b, \boldsymbol{\xi}} \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{N} \xi_i \quad \text{subject to} \quad y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0. \tag{14.15}$$

A large C penalizes total violations more strongly relative to $\|\boldsymbol{w}\|^2/2$; it does not guarantee zero slack in a nonseparable data set.

Feasibility and attainment. We should first verify that introducing slacks resolves feasibility and still permits an optimal solution. The objective is the sum of a convex quadratic and a linear function, while all constraints are affine. The model is therefore convex. Substituting $(\boldsymbol{w}, b, \boldsymbol{\xi}) = (\mathbf{0}, 0, \mathbf{1})$ satisfies every constraint, irrespective of the labels, so its feasible set is nonempty.

For attainment, take a feasible minimizing sequence with objective values at most some finite M. Since $C > 0$ and every $\xi_i \geq 0$, we have

$$\tfrac{1}{2}\left\|\boldsymbol{w}^{(k)}\right\|^2 \leq M, \qquad 0 \leq \xi_i^{(k)} \leq M/C \quad (i = 1{:}N).$$

Hence $\boldsymbol{w}^{(k)}$ and $\boldsymbol{\xi}^{(k)}$ are bounded. If both labels occur, fix indices i_+ and i_- with these two labels. Their constraints give

$$1 - \xi_{i_+}^{(k)} - \boldsymbol{w}^{(k)\mathsf{T}}\boldsymbol{x}_{i_+} \leq b^{(k)} \leq -1 + \xi_{i_-}^{(k)} - \boldsymbol{w}^{(k)\mathsf{T}}\boldsymbol{x}_{i_-}.$$

Both bounds are bounded because $\boldsymbol{w}^{(k)}$ and the slacks are bounded, so $b^{(k)}$ is bounded as well. A convergent subsequence has a feasible limit, since the constraints are closed, and continuity of the objective makes that limit a minimizer. If only one label occurs, choose $\boldsymbol{w} = \mathbf{0}$, $\boldsymbol{\xi} = \mathbf{0}$, and $b = 1$ for the positive class or $b = -1$ for the negative class. This feasible point has value zero, the lower bound on the objective. In either case, the soft-margin model attains its minimum.

Eliminating the slacks. We can express the same optimization using only the separating variables by minimizing over the slacks first. Fix $(\boldsymbol{w}, b)$. Feasibility requires

$$\xi_i \geq 0, \qquad \xi_i \geq 1 - y_i(\boldsymbol{w}^{\mathsf{T}}\boldsymbol{x}_i + b).$$

Because $C > 0$, the objective is strictly increasing in each ξ_i. Its minimizing feasible value is therefore $\xi_i = \max\{0, 1 - y_i(\boldsymbol{w}^{\mathsf{T}}\boldsymbol{x}_i + b)\}$. Every optimum must use this choice: reducing any larger slack would keep feasibility and strictly decrease the objective. Substituting the minimizing slacks yields

$$\min_{\boldsymbol{w},b} \ \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{N} \max\{0, 1 - y_i(\boldsymbol{w}^{\mathsf{T}}\boldsymbol{x}_i + b)\}. \tag{14.16}$$

Conversely, adjoining these slacks to any $(\boldsymbol{w}, b)$ gives a feasible point of Equation (14.15) with exactly the displayed value. The two formulations are therefore equivalent. The function $\max\{0, 1 - y_i s_i\}$ is the hinge penalty on the signed response.

The soft-margin KKT system. To interpret the soft-margin solution through multipliers, return to the smooth slack-variable formulation. Its constraints are affine, so the attained minimizer admits KKT multipliers by Theorem 14.2, even without LICQ. Introduce multipliers $\nu_i \geq 0$ for $1 - \xi_i - y_i(\boldsymbol{w}^{\mathsf{T}}\boldsymbol{x}_i + b) \leq 0$ and $\rho_i \geq 0$ for $-\xi_i \leq 0$. The Lagrangian is

$$L = \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_i \xi_i + \sum_i \nu_i[1 - \xi_i - y_i(\boldsymbol{w}^{\mathsf{T}}\boldsymbol{x}_i + b)] - \sum_i \rho_i\xi_i.$$

Differentiating with respect to $\boldsymbol{w}$, b, and ξ_i gives

$$\nabla_{\boldsymbol{w}} L = \boldsymbol{w} - \sum_i \nu_i y_i \boldsymbol{x}_i, \qquad \frac{\partial L}{\partial b} = -\sum_i \nu_i y_i, \qquad \frac{\partial L}{\partial \xi_i} = C - \nu_i - \rho_i.$$

Set these derivatives to zero and append dual feasibility and complementary slackness. At an optimum the resulting equations are

$$\begin{aligned} \boldsymbol{w}^* &= \sum_i \nu_i^* y_i \boldsymbol{x}_i, & \sum_i \nu_i^* y_i &= 0, \\ C - \nu_i^* - \rho_i^* &= 0, & \nu_i^*, \rho_i^* &\geq 0, \\ \nu_i^*[1 - \xi_i^* - y_i({\boldsymbol{w}^*}^{\mathsf{T}} \boldsymbol{x}_i + b^*)] &= 0, & \rho_i^* \xi_i^* &= 0, \end{aligned}$$

together with primal feasibility. Conversely, convex KKT sufficiency applies to any feasible point and multipliers satisfying this system. We have therefore obtained necessary and sufficient conditions for optimality.

To read the equations geometrically, first solve the slack stationarity equation for ρ_i^*: it gives $\rho_i^* = C - \nu_i^*$. The two multiplier signs therefore restrict ν_i^* to $[0, C]$. We can now inspect the three parts of this interval.

1. If $\nu_i^* = 0$, then $\rho_i^* = C > 0$. Complementarity $\rho_i^* \xi_i^* = 0$ forces $\xi_i^* = 0$, and primal feasibility then gives $y_i({\boldsymbol{w}^*}^{\mathsf{T}} \boldsymbol{x}_i + b^*) \geq 1$. Such an observation is on or outside its correct margin.
2. If $0 < \nu_i^* < C$, then $\rho_i^* = C - \nu_i^* > 0$, so again $\xi_i^* = 0$. This time $\nu_i^* > 0$ also makes the margin constraint active. Hence $y_i({\boldsymbol{w}^*}^{\mathsf{T}} \boldsymbol{x}_i + b^*) = 1$: the observation lies exactly on its margin plane.
3. If $\nu_i^* = C$, then $\nu_i^* > 0$ because $C > 0$. The margin constraint is active, giving signed response $1 - \xi_i^* \leq 1$. The slack may be zero or positive, so this case includes a margin point, a correctly classified point inside the margin, a point on the decision plane, and a misclassified point, according to the value of ξ_i^*.

We can also start from a positive slack. If $\xi_i^* > 0$, complementarity forces $\rho_i^* = 0$; substituting into slack stationarity gives $\nu_i^* = C$. Thus every positive slack is associated with a multiplier at the upper bound.

Warning

The bound $\nu_i = C$ does not by itself distinguish a point exactly on the margin from one inside the margin or misclassified; the slack and signed response must also be examined. Likewise, $\nu_i = 0$ need not imply strict margin in a degenerate data set.

Warning

To recover the soft-margin intercept from an active observation, solve $y_i(\boldsymbol{w}^{\mathsf{T}} \boldsymbol{x}_i + b) = 1 - \xi_i$ for b. This gives $b = y_i(1 - \xi_i) - \boldsymbol{w}^{\mathsf{T}} \boldsymbol{x}_i$. It reduces to $b = y_i - \boldsymbol{w}^{\mathsf{T}} \boldsymbol{x}_i$ when the slack is zero, in particular when $0 < \nu_i < C$. A loaded observation with $\nu_i = C$ may have positive slack, so the hard-margin formula cannot be applied to every loaded observation. There need not be a free-margin index with $0 < \nu_i < C$.
To test whether the intercept must be unique, take two coincident observations $\boldsymbol{x}_+ = \boldsymbol{x}_- = \boldsymbol{0}$ with opposite labels. Their scores do not depend on $\boldsymbol{w}$, so minimizing $\|\boldsymbol{w}\|^2/2$

selects $\boldsymbol{w} = \mathbf{0}$. For arbitrary b, the minimal slack sum is $\max\{0, 1-b\}+\max\{0, 1+b\} \geq 2$. If $b \in [-1, 1]$, both terms are nonnegative before taking the maxima and their sum is exactly 2. Thus every such b, with $\xi_+ = 1 - b$ and $\xi_- = 1 + b$, attains value $2C$. Unlike the two-class hard-margin case, a soft-margin intercept can therefore be nonunique.

Chapter summary

- The canonical smooth convex problem has convex f, h_j and affine g_i. These assumptions are sufficient for a convex feasible set.
- The feasible-set variational inequality is equivalent to global optimality and, by convexity of the set, to the tangent-cone inequality.
- KKT first-order conditions are equivalent to the linearized-cone inequality. Every convex KKT point is globally optimal without any qualification.
- LICQ gives the reverse implication at an attained minimizer, and unique multipliers. It does not guarantee primal attainment. Affine constraints give multiplier necessity without LICQ.
- Strict convexity gives at most one minimizer but does not imply SOSC. SOSC gives strict local minimality and, in a convex problem, global uniqueness.
- In hard-margin SVM, active points are geometric support vectors and positive-multiplier points are loaded. Only loaded points determine $\boldsymbol{w}$; any active margin point gives the same intercept in exact arithmetic.
- With both classes and feasible hard margins, the normalized separating pair $(\boldsymbol{w}^*, b^*)$ is unique even though multipliers may not be.
- Soft margins always give a feasible convex model; slacks measure functional margin deficits and $0 \leq \nu_i \leq C$. Feasibility does not mean perfect linear classification of nonseparable data.

14.7 Exercises

Exercise 14.1 (Convex KKT certificate)**.** Solve

$$\min_{x_1,x_2}(x_1 - 3)^2 + 2(x_2 + 1)^2 \quad \text{subject to} \quad 2x_1 + x_2 \leq 1$$

using KKT, and explain exactly where convexity upgrades the candidate to a global solution.

Exercise 14.2 (Nonunique convex solutions)**.** Find the complete solution set and all KKT multipliers of

$$\min_{x_1,x_2}(x_1 + x_2 - 1)^2 \quad \text{subject to} \quad -x_1 \leq 0, \quad -x_2 \leq 0, \quad x_1 + x_2 - 2 \leq 0.$$

Identify why convexity does not imply uniqueness in this example.

Exercise 14.3 (A change of equality representation)**.** Compare the two representations of the same convex feasible set

$$\min_{x_1,x_2}(x_1-2)^2+x_2 \quad \text{subject to} \quad \begin{cases} g(\boldsymbol{x})=x_2=0, & \text{representation A,} \\ \widetilde{g}(\boldsymbol{x})=x_2^2=0, & \text{representation B.} \end{cases}$$

Find the global minimizer. For each representation, determine whether KKT multipliers exist and whether LICQ holds. Explain why the convex KKT theorem applies directly to representation A but not B, despite identical feasible sets.

Exercise 14.4 (Dependent affine constraints)**.** Solve

$$\min_{x_1,x_2} \tfrac{1}{2}(2x_1^2+3x_2^2) \quad \text{subject to} \quad 3-2x_1+x_2\le 0, \quad 6-4x_1+2x_2\le 0.$$

Show that LICQ fails at the optimizer, find the entire set of KKT multipliers, and explain why Theorem 14.2 still applies.

Exercise 14.5 (A hand-solvable hard-margin SVM)**.** For the one-dimensional data $(-3,-1)$, $(-1,-1)$, $(2,+1)$, and $(4,+1)$, solve the hard-margin SVM. Determine w^*, b^*, the margin width, and one valid multiplier vector. Identify the support vectors.

Exercise 14.6 (Soft-margin KKT classification)**.** At a soft-margin KKT point, suppose the signed responses of four observations are $3/2$, 1, $1/2$, and $-1/4$. Determine what can and cannot be concluded about each ξ_i, ν_i, and ρ_i. State explicitly where degeneracy prevents a strict conclusion.

Exercise 14.7 (Degenerate support vectors)**.** Construct a separable two-dimensional data set with exactly three observations, all on its two optimal margin lines, such that LICQ holds but one observation has a zero KKT multiplier. Verify the full KKT system and explain why active contact need not be loaded even with unique multipliers.